Solutions to the Self-Paced, Proof-Level Syllabus

for *"The Noisy Folded Limit Cycle: Tracy–Widom Statistics of Canard Escape"*

This document works every *milestone problem* (Part I, by module) and then answers the thirteen *"Can you explain it?"* self-test questions of §11 (Part II). Throughout, the inner equation is the swept Riccati

$$\mathrm{d}R = (R^2 - Y)\,\mathrm{d}T + \eta\,\mathrm{d}B, \qquad Y = Y_0 - T, \qquad \eta = \sigma/\sqrt{\varepsilon_2}, \tag{1}$$

with deterministic limit the Airy equation and Cole–Hopf linearisation $R = -u'/u$, $u'' = (Y - \eta\xi)u$. The dictionary to the stochastic Airy operator is $\eta = 2/\sqrt{\beta}$, $\beta = 4/\eta^2$, and the central result is $Y_{\text{node}} \overset{d}{=} -\Lambda_0(\beta) \overset{d}{=} \mathrm{TW}_\beta$.

Part I — Milestone problems

Module F1 — ODEs, invariant manifolds, bifurcations

F1.1 (Saddle–node normal form via centre manifold).

Solution. Let $\dot{x} = f(x, y, \mu)$, $\dot{y} = g(x, y, \mu)$ be a planar family with f, g smooth, an equilibrium at the origin for $\mu = 0$, and linearisation having a simple zero eigenvalue (eigendirection x) and one eigenvalue $-\lambda < 0$ (direction y). The Centre Manifold Theorem gives a locally invariant C^k curve $y = h(x, \mu)$, $h = O(x^2, \mu)$, attracting at rate λ; the dynamics reduce to the scalar field $\dot{x} = F(x, \mu) := f(x, h(x, \mu), \mu)$. Expand $F(x, \mu) = a\mu + \frac{1}{2}b\,x^2 + O(\mu x, \mu^2, x^3)$ with $a = \partial_\mu F(0, 0) = f_\mu$ and $b = \partial_x^2 F(0, 0) = f_{xx}$ (the genericity/transversality conditions are $a \neq 0$, $b \neq 0$). The near-identity rescaling $x \mapsto |b/2|^{-1}x$, $\mu \mapsto (a)\,\mu$ and a time reparametrisation put this in the universal form

$$\dot{x} = \mu \pm x^2, \qquad \text{sign} = \text{sign}(b).$$

Two equilibria $x = \pm\sqrt{\mp\mu}$ exist on one side of $\mu = 0$ and collide and disappear at $\mu = 0$: the saddle–node.

F1.2 (Fold of cycles = saddle–node of the Poincaré map).

Solution. Take a cross-section Σ transverse to a limit cycle Γ and let $P : \Sigma \to \Sigma$ be the first-return (Poincaré) map; $\Gamma \cap \Sigma$ is a fixed point $p_* = P(p_*)$, and the nontrivial Floquet multipliers of Γ are the eigenvalues of $DP(p_*)$. An attracting and a saddle cycle correspond to two fixed points of P; a *fold of limit cycles* is their collision and annihilation, i.e. a *saddle–node bifurcation of the fixed point of* P. At such a bifurcation $DP(p_*)$ has an eigenvalue equal to $+1$ (a one-dimensional centre direction of the map, $\det(DP - I) = 0$ with the eigenvalue crossing through $+1$), while the trivial multiplier along the flow is unchanged. Equivalently, one nontrivial Floquet multiplier of the cycle crosses $+1$ — the periodic-orbit signature of the fold.

F1.3 (The deterministic normal form; positivity and periodicity of a, b, c).

Solution. After reduction to the centre manifold of the folded cycle manifold and introduction of the fast phase θ (period normalised to 1), the deterministic skeleton is the *fold family*

$$r' = b(\theta)\,r^2 - a(\theta)\,y, \qquad y' = -\,c(\theta)\,\varepsilon_2, \qquad \theta' = 1,$$

i.e. a θ-parametrised quadratic fold with a slow drift of the bifurcation variable y. The coefficients a, b, c are geometric quantities (curvature of the cycle manifold, fold coefficient, slow-drift rate) *evaluated along the underlying periodic orbit*; since that orbit has period 1 in θ, a, b, c are 1-periodic functions of θ. They are *positive* because: $b > 0$ encodes a genuine quadratic (non-degenerate) fold; $a > 0$ that the attracting branch lies on the correct side ($r \sim -\sqrt{ay/b}$ is real and attracting for $y > 0$); and $c > 0$ that the slow variable is swept monotonically through the fold. Their combination $G = \sqrt{ac/b}$ is the geometry factor in the physical critical noise $\sigma_* = 2\sqrt{\pi}\,\sqrt{\varepsilon_2}\,G$.

Module F2 — Measure-theoretic probability

F2.1 (Lévy continuity: convergence in distribution ⇔ pointwise convergence of characteristic functions).

Solution. Let $\varphi_n(t) = \mathbb{E}e^{itX_n}$. ($\Rightarrow$) If $X_n \Rightarrow X$ then, since $x \mapsto e^{itx}$ is bounded and continuous, $\varphi_n(t) \to \varphi(t)$ for every t by the definition of weak convergence. ($\Leftarrow$) Suppose $\varphi_n(t) \to \varphi(t)$ pointwise with φ continuous at 0. The bound $\mathbb{P}(|X_n| > 2/\delta) \le \frac{1}{\delta}\int_{-\delta}^{\delta}\big(1 - \operatorname{Re}\varphi_n(t)\big)\mathrm{d}t$ together with $\varphi(0) = 1$ and continuity of φ at 0 gives, via dominated convergence, $\limsup_n \mathbb{P}(|X_n| > 2/\delta) \le \frac{1}{\delta}\int_{-\delta}^{\delta}(1 - \operatorname{Re}\varphi) \to 0$ as $\delta \to 0$; hence $\{X_n\}$ is tight. By Prokhorov every subsequence has a further weakly convergent subsequence, whose limit has characteristic function φ (by the forward direction); since φ determines the law uniquely (inversion), all subsequential limits coincide, so $X_n \Rightarrow X$ with $\mathbb{E}e^{itX} = \varphi(t)$.

F2.2 (Brownian stationarity, reflection, and white noise).

Solution. Let B be standard Brownian motion. *Stationary increments:* for $s < t$, $B_t - B_s \sim N(0, t-s)$ depends only on $t - s$, and increments over disjoint intervals are independent (both from the defining Gaussian covariance $\mathbb{E}B_sB_t = s \wedge t$). *Reflection $t \mapsto -t$:* on a symmetric interval define $\tilde{B}_t := B_{-t}$ (or, for two-sided white-noise purposes, note $-B$ and the time-reversed process are again Brownian); the covariance is preserved and Gaussianity is preserved, so the law is invariant under reflection. *White noise:* $\xi := \mathrm{d}B/\mathrm{d}t$ is defined as the distributional (Schwartz) derivative; for test functions ϕ, $\langle \xi, \phi\rangle := -\int B\,\phi'\,\mathrm{d}t = \int \phi\,\mathrm{d}B$, a centred Gaussian field with $\mathbb{E}\langle\xi,\phi\rangle\langle\xi,\psi\rangle = \int \phi\psi$, i.e. a *stationary* generalised process with covariance "$\delta(t-s)$", invariant under translation and reflection. These two invariances are exactly what collapses the random spectral level in §4.3 to a deterministic one.

Module F3 — Functional analysis, Sturm–Liouville/oscillation

F3.1 (Variational characterisation of the lowest Dirichlet eigenvalue).

Solution. For $H = -\partial_x^2 + V$ on $L^2(0,\infty)$ with Dirichlet data $\psi(0) = 0$ and V bounded below and confining, H is self-adjoint and bounded below with form $\mathcal{Q}[\psi] = \int_0^\infty(\psi'^2 + V\psi^2)$ on the form domain H_0^1. The spectral theorem gives, for the bottom of the spectrum,

$$\Lambda_0 = \inf_{\psi\in H_0^1,\ \|\psi\|=1} \mathcal{Q}[\psi] = \inf_{\psi\neq 0}\frac{\int(\psi'^2 + V\psi^2)}{\int\psi^2}.$$

($\le$) Any admissible ψ gives an upper bound (Rayleigh quotient). ($\ge$) Diagonalise: $\psi = \sum c_k\phi_k$, $H\phi_k = \lambda_k\phi_k$, $\mathcal{Q}[\psi] = \sum\lambda_k|c_k|^2 \ge \Lambda_0\sum|c_k|^2 = \Lambda_0\|\psi\|^2$. The infimum is attained at the ground state, which (Perron–Frobenius/positivity) may be taken nodeless on $(0,\infty)$.

F3.2 (Dirichlet domain monotonicity).

Solution. Let $\Omega_1 \subset \Omega_2$ and $\Lambda_0(\Omega_i)$ the lowest Dirichlet eigenvalue of $-\partial_x^2 + V$ on Ω_i. Extension by zero embeds $H_0^1(\Omega_1) \hookrightarrow H_0^1(\Omega_2)$ isometrically for both $\int\psi^2$ and $\mathcal{Q}[\psi]$ (the extended function is H^1 because it vanishes on $\partial\Omega_1$). Hence the minimising set for Ω_2 contains that for Ω_1, and taking an infimum over a *larger* set can only decrease it:

$$\Lambda_0(\Omega_2) = \inf_{H_0^1(\Omega_2)} \frac{\mathcal{Q}}{\|\cdot\|^2} \le \inf_{H_0^1(\Omega_1)} \frac{\mathcal{Q}}{\|\cdot\|^2} = \Lambda_0(\Omega_1).$$

So shrinking the domain raises the ground state. This monotonicity is the engine of the shooting argument: it makes the spectral function $G(\ell)$ strictly increasing.

F3.3 (Sturm fact: $Y_{\mathrm{node}} = a_1 = -\Lambda_0(\infty)$ for the deterministic Airy operator).

Solution. Consider $H = -\partial_x^2 + x$ on $[\ell, \infty)$ with Dirichlet data at ℓ. By Sturm oscillation theory the number of nodes in (ℓ, ∞) of the recessive (decaying) solution at energy 0 equals the number of eigenvalues of H below 0. The recessive solution is $\mathrm{Ai}(x)$ (up to scale), whose first zero is $a_1 \approx -2.3381$. Sweeping the left endpoint: $G(\ell) := \Lambda_0\big(H|_{[\ell,\infty)}\big)$ is strictly increasing (F3.2) and $G(\ell) = 0 \iff$ Ai has its first node exactly at the energy-zero turning configuration with left endpoint $\ell = a_1$. Equivalently, on the whole line the deterministic peel-off ($R = \mathrm{Ai}'/\mathrm{Ai} \to \infty$) occurs at $Y = a_1$, and $a_1 = -\Lambda_0(\infty)$ where $\Lambda_0(\infty)$ is the bottom eigenvalue of $-\partial_x^2 + x$ on $[0, \infty)$. Thus $Y_{\mathrm{node}} = a_1 = -\Lambda_0(\infty)$ — the deterministic ($\beta \to \infty$) limit of the main theorem.

Module F4 — Airy, Riccati, Cole–Hopf, asymptotics

F4.1 ($R = \mathrm{Ai}'/\mathrm{Ai}$ solves the Riccati and escapes at a_1).

Solution. Write $R(Y) = \mathrm{Ai}'(Y)/\mathrm{Ai}(Y)$. Then

$$\frac{\mathrm{d}R}{\mathrm{d}Y} = \frac{\mathrm{Ai}''\mathrm{Ai} - \mathrm{Ai}'^2}{\mathrm{Ai}^2} = \frac{\mathrm{Ai}''}{\mathrm{Ai}} - R^2 = Y - R^2,$$

using the Airy equation $\mathrm{Ai}''(Y) = Y\,\mathrm{Ai}(Y)$. In the inner time T with $Y = Y_0 - T$ ($\mathrm{d}Y/\mathrm{d}T = -1$) this is $\mathrm{d}R/\mathrm{d}T = -(Y - R^2) = R^2 - Y$, i.e. the deterministic Riccati (1). As T increases Y decreases from $Y_0 > 0$; on the attracting branch $R \sim -\sqrt{Y}$ (this is the canard, the recessive Ai solution). R blows up exactly where $\mathrm{Ai}(Y) = 0$; the first such Y (reached as Y decreases) is $a_1 \approx -2.3381$, where $\mathrm{Ai}'(a_1) > 0$ so $R \to +\infty$. This is the deterministic canard escape.

F4.2 (Deterministic Cole–Hopf).

Solution. Put $R = -u'/u$. Then $R' = -\dfrac{u''}{u} + \left(\dfrac{u'}{u}\right)^2 = -\dfrac{u''}{u} + R^2$, so $R' = R^2 - Y$ is equivalent to $u'' = Y\,u$ (the linear Airy/Schrödinger equation). A finite-time blow-up of $R = -u'/u$ occurs precisely where $u = 0$ with $u' \neq 0$, i.e. at a simple zero (node) of u. Hence "Riccati escape" = "first node of u" — the algebraic skeleton of Lemma 6, which survives verbatim when $Y \mapsto Y - \eta\xi$.

F4.3 (Watson/Laplace evaluation of the hazard integral).

Solution. With $I = \displaystyle\int_0^\infty Y^{1/2}\, e^{-8Y^{3/2}/(3\eta^2)}\,\mathrm{d}Y$, substitute $w = Y^{3/2}$, $\mathrm{d}w = \frac{3}{2}Y^{1/2}\mathrm{d}Y$, so $Y^{1/2}\mathrm{d}Y = \frac{2}{3}\mathrm{d}w$:

$$I = \frac{2}{3}\int_0^\infty e^{-8w/(3\eta^2)}\,\mathrm{d}w = \frac{2}{3}\cdot\frac{3\eta^2}{8} = \frac{\eta^2}{4}.$$

(The substitution makes the exponent linear, so Watson's lemma is exact here.) Therefore the integrated hazard is $H(\eta) = \frac{1}{\pi}I = \eta^2/4\pi$, and the critical level $H = 1$ gives $\eta_* = 2\sqrt{\pi}$.

Module A — GSPT, fold/canard, geometric blow-up

A.1 (K_2 rescaling → inner Riccati; $\eta = \sigma/\sqrt{\varepsilon_2}$).

Solution. Insert the rescaling chart $r = \varepsilon_2^{1/3}R$, $y = \varepsilon_2^{2/3}Y$, $T = \varepsilon_2^{1/3}t$ (so $\mathrm{d}T = \varepsilon_2^{1/3}\mathrm{d}t$) into the noisy normal form $\mathrm{d}r = (br^2 - ay)\mathrm{d}t + \sigma\,\mathrm{d}W$, $\mathrm{d}y = -c\,\varepsilon_2\,\mathrm{d}t$. Drift: $\mathrm{d}r = \varepsilon_2^{1/3}\mathrm{d}R$, while $(br^2 - ay)\mathrm{d}t = (b\varepsilon_2^{2/3}R^2 - a\varepsilon_2^{2/3}Y)\varepsilon_2^{-1/3}\mathrm{d}T = \varepsilon_2^{1/3}(bR^2 - aY)\mathrm{d}T$; absorbing a, b into the unit normalisation gives $\mathrm{d}R = (R^2 - Y)\mathrm{d}T$. Noise: a Brownian increment in t scales as $\mathrm{d}W_t = \varepsilon_2^{-1/6}\mathrm{d}W_T$ (since $\mathrm{d}t = \varepsilon_2^{-1/3}\mathrm{d}T$ and $\mathrm{d}W \sim \sqrt{\mathrm{d}t}$), so the noise term is $\sigma\varepsilon_2^{-1/6}\mathrm{d}W_T$; dividing the whole line by $\varepsilon_2^{1/3}$ leaves

$$\mathrm{d}R = (R^2 - Y)\mathrm{d}T + \sigma\,\varepsilon_2^{-1/6-1/3}\,\mathrm{d}W_T = (R^2 - Y)\mathrm{d}T + \eta\,\mathrm{d}B, \qquad \eta = \sigma\,\varepsilon_2^{-1/2} = \sigma/\sqrt{\varepsilon_2}.$$

A.2 ($(1, 2, 3)$ weights; what each chart resolves).

Solution. Assign weights $(p_r, p_y, p_\varepsilon)$ to (r, y, ε_2) and demand the fold normal form $r' = r^2 - y$, $\dot{\varepsilon_2} = 0$ (with slow drift $y' \sim \varepsilon_2$) be invariant under $(r, y, \varepsilon_2) \mapsto (\kappa^{p_r}r, \kappa^{p_y}y, \kappa^{p_\varepsilon}\varepsilon_2)$ after a time rescaling $t \mapsto \kappa^{-p_r}t$. Matching $r' = r^2$: $p_r - (-p_r) = 2p_r$ is automatic; matching r^2 vs y forces $p_y = 2p_r$; matching the slow equation $y' \sim \varepsilon_2$ forces $p_\varepsilon = p_y + p_r = 3p_r$. Normalising $p_r = 1$ gives the weights $(1, 2, 3)$ on (r, y, ε_2). The blow-up $(r, y, \varepsilon_2) = (\bar{r}\rho, \bar{y}\rho^2, \bar{\varepsilon_2}\rho^3)$ desingularises the fold; the three charts resolve: K_1 (entry) the incoming attracting Fenichel manifold; K_2 (rescaling) the inner Airy/Riccati dynamics where hyperbolicity is fully lost; K_3 (exit) the departure onto the fast fibre and matching to the outer solution.

A.3 (Canard and escape in the inner equation).

Solution. The deterministic inner equation is $\mathrm{d}R/\mathrm{d}T = R^2 - Y$, $Y = Y_0 - T$; as in F4.1 its canard is $R = \mathrm{Ai}'(Y)/\mathrm{Ai}(Y)$, the solution tracking the attracting branch $R \sim -\sqrt{Y}$ for $Y \gg 0$ and escaping to $+\infty$ at the first Airy zero $Y = a_1 \approx -2.3381$. This is the deterministic peel-off the noise perturbs.

Module B1 — Brownian motion, Itô, SDEs

B1.1 (Lemma 6: Cole–Hopf with no Itô correction).

Solution. Let $\mathrm{d}u = v\,\mathrm{d}T$, $\mathrm{d}v = (Yu)\,\mathrm{d}T - \eta u\,\mathrm{d}B$, and $R = -v/u$. Only v carries noise, so $(\mathrm{d}u)^2 = \mathrm{d}u\,\mathrm{d}v = 0$ and $(\mathrm{d}v)^2 = \eta^2u^2\,\mathrm{d}T$. With $R = R(u, v) = -v/u$, $R_u = v/u^2$, $R_v = -1/u$, $R_{vv} = 0$, $R_{uv} = 1/u^2$, $R_{uu} = -2v/u^3$. Itô:

$$\mathrm{d}R = R_u\,\mathrm{d}u + R_v\,\mathrm{d}v + \tfrac{1}{2}R_{uu}(\mathrm{d}u)^2 + R_{uv}\,\mathrm{d}u\,\mathrm{d}v + \tfrac{1}{2}R_{vv}(\mathrm{d}v)^2 = \frac{v}{u^2}(v\,\mathrm{d}T) - \frac{1}{u}\big[(Yu)\mathrm{d}T - \eta u\,\mathrm{d}B\big].$$

The second-order terms vanish because $(\mathrm{d}u)^2 = \mathrm{d}u\,\mathrm{d}v = 0$ and $R_{vv} = 0$. Hence $\mathrm{d}R = \frac{v^2}{u^2}\mathrm{d}T - Y\,\mathrm{d}T + \eta\,\mathrm{d}B = (R^2 - Y)\,\mathrm{d}T + \eta\,\mathrm{d}B$ — *exactly* the Riccati, with no Itô–Stratonovich correction. (Equivalently the diffusion vector $g = (0, -\eta u)$ has $\frac{1}{2}(g\cdot\nabla)g = 0$.)

B1.2 (Proposition 8: change of variables is exact).

Solution. Combine A.1 and B1.1. Apply the deterministic chart map (3) to the physical SDE (2); the drift transforms as in A.1 to $(R^2 - Y)\mathrm{d}T$, and because the physical noise is *additive* (constant coefficient σ), the Itô term transforms with no correction (B1.1), giving $\eta = \sigma/\sqrt{\varepsilon_2}$ exactly — not merely to leading order. Thus the blown-up, rescaled physical system *is* the canonical noisy Riccati (1).

B1.3 (Peel-off scalings $r_{\rm peel} \sim \sigma^{2/3},\ y_{\rm peel} \sim \sigma^{4/3}$).

Solution. Escape is an $O(1)$ event in inner variables ($R, Y = O(1)$) when $\eta = \sigma/\sqrt{\varepsilon_2} = O(1)$, i.e. at criticality $\varepsilon_2 \sim \sigma^2$. Undoing the chart, $r = \varepsilon_2^{1/3} R \sim (\sigma^2)^{1/3} = \sigma^{2/3}$ and $y = \varepsilon_2^{2/3} Y \sim (\sigma^2)^{2/3} = \sigma^{4/3}$. These are the physical noise-induced peel-off scales.

Module B2 — Berglund–Gentz sample-path tubes

B2.1 (Integrated Kramers hazard).

Solution. The instantaneous separatrix-crossing rate near the fold is Arrhenius in the local barrier $\propto Y^{3/2}$; the integrated hazard is $H(\eta) = \frac{1}{\pi}\int_0^\infty Y^{1/2} e^{-8Y^{3/2}/3\eta^2}\,\mathrm{d}Y = \eta^2/4\pi$ by F4.3. Setting $H = 1$ gives the critical effective noise $\eta_* = 2\sqrt{\pi}$ (hence physically $\sigma_* = 2\sqrt{\pi}\sqrt{\varepsilon_2}\,G$). Note H is the saddle-crossing hazard and is *polynomial* in η, distinct from the exponential canonical escape probability of §5.

B2.2 (Telescoping the transition-error recursion).

Solution. Write the transverse deviation after chart i as $|\delta_i| \le L_i|\delta_{i-1}| + |\zeta_i|$, L_i the local contraction/expansion factor and ζ_i the fresh fluctuation. Iterating,

$$|\delta_n| \le \Big(\textstyle\prod_{i=1}^n L_i\Big)|\delta_0| + \sum_{k=1}^n \Big(\prod_{i=k+1}^n L_i\Big)|\zeta_k|.$$

At the fold the dynamics are exponentially contracting along the attracting branch, $L_{\rm fold} \ll 1$, so the prefactor $\prod L_i$ multiplying the *entry* error δ_0 is exponentially small: the incoming fluctuation is forgotten, and the exit law is set only by the local fluctuations ζ_k accumulated inside K_2. This is the renewal/"forgetting" that lets the inner problem be analysed in isolation (Lemma 7, §6.1).

Module B3 — Large deviations: Schilder, contraction

B3.1 ($\Lambda_0(\eta)$ as a continuous functional of the path).

Solution. By Rayleigh–Ritz the ground state of $M = -\partial_x^2 + x - \eta\dot W$ on $[0,\infty)$ (Dirichlet) is

$$\Lambda_0(\eta) = \inf_{\|\psi\|=1,\ \psi(0)=0}\Big(K[\psi] + \eta N[\psi]\Big), \quad K[\psi] = \int_0^\infty (\psi'^2 + x\psi^2), \quad N[\psi] = \int_0^\infty \psi^2\,\mathrm{d}W.$$

Integrating by parts (and using $\psi(0) = 0$), $N[\psi] = -2\int_0^\infty \psi\psi'\,W\,\mathrm{d}x$, which is a *bounded continuous* linear functional of the path $W \in C_0$. Hence $\Lambda_0(\eta) = F(\eta W)$ for the continuous map $F : C_0 \to \mathbb{R}$, $F(h) = \inf_\psi(K[\psi] - 2\int \psi\psi' h)$ (an infimum of continuous affine functionals, hence upper semicontinuous; continuity holds on the relevant compact sublevel sets).

B3.2 (Schilder + contraction $\Rightarrow$ the rate constant c).

Solution. Schilder: ηW satisfies an LDP with good rate $I(h) = \frac{1}{2}\int h'^2$ as $\eta \to 0$. The early-escape event is $\{\Lambda_0(\eta) < 0\} = \{F(\eta W) < 0\}$; by the contraction principle its rate is

$$c = \inf\Big\{\tfrac{1}{2}\textstyle\int g'^2 :\ F(g) \le 0\Big\} = \tfrac{1}{2}\,\min_\psi \frac{K[\psi]^2}{\int \psi^4},$$

the last equality by optimising the linear-in-g constraint (the worst-case path aligned with ψ^2). The Euler–Lagrange equation of $\min K[\psi]^2/\int\psi^4$ is, after scaling,

$$-g'' + xg = g^3,$$

the nonlinear-Airy ground state; then $P_{\rm early} = \exp(-c/\eta^2(1 + o(1)))$ with $c = \int g'^2$.

Module C — Random matrices and the stochastic Airy operator

C.1 (RRV Riccati diffusion = the swept inner Riccati).

Solution. Ramírez–Rider–Virág characterise the ground state $\Lambda_0(\beta)$ of $\mathcal{H}_\beta = -\partial_x^2 + x + \frac{2}{\sqrt{\beta}}b'$ by the explosion of the Riccati diffusion $\mathrm{d}p = (x - \Lambda - p^2)\,\mathrm{d}x + \frac{2}{\sqrt{\beta}}\,\mathrm{d}b$: Λ is an eigenvalue iff p explodes to $-\infty$. Term-by-term this is the swept inner Riccati (1) (with $x \leftrightarrow$ swept level, $p \leftrightarrow R$ up to orientation) provided the noise strengths match: $\frac{2}{\sqrt{\beta}} = \eta$, i.e. $\eta = 2/\sqrt{\beta}$, $\beta = 4/\eta^2$. Hence the paper's inner equation *is* RRV's diffusion, and $Y_{\mathrm{node}} \overset{d}{=} -\Lambda_0(\beta) \overset{d}{=} \mathrm{TW}_\beta$.

C.2 ($\beta \to \infty$ recovers the deterministic Airy zero).

Solution. As $\beta \to \infty$ ($\eta \to 0$) the noise $\frac{2}{\sqrt{\beta}}b' \to 0$ and $\mathcal{H}_\beta \to -\partial_x^2 + x$, the deterministic Airy operator, whose ground state is $-a_1$. Thus $-\Lambda_0(\infty) = a_1 \approx -2.3381$, recovering F3.3/F4.1: the noiseless peel-off is the first Airy zero. $\mathrm{TW}_\beta \to \delta_{a_1}$ (fluctuations of order η shrink to zero).

C.3 ($\beta = 1, 2, 4$ and non-integer β).

Solution. $\beta = 1, 2, 4$ are the Dyson symmetry classes GOE/GUE/GSE (real/complex/quaternion entries), whose soft-edge fluctuations are $\mathrm{TW}_1, \mathrm{TW}_2, \mathrm{TW}_4$. The stochastic-Airy / β-ensemble construction extends TW_β to *all* $\beta > 0$ via the operator $\mathcal{H}_\beta$. The paper realises *continuous, generally non-integer* $\beta = 4/\eta^2$ tuned by the physical noise-to-curvature ratio η — e.g. $\eta = \sqrt{2} \Rightarrow \beta = 2$, $\eta = 2 \Rightarrow \beta = 1$ — so the canard-escape law sweeps the whole Tracy–Widom family.

Module D — Painlevé II and the exact laws

D.1 ($g = \sqrt{2}\,s$ maps the nonlinear Airy ground state to Painlevé II).

Solution. In $-g'' + xg = g^3$ put $g = \sqrt{2}\,s$: $-\sqrt{2}\,s'' + \sqrt{2}\,xs = 2\sqrt{2}\,s^3$; divide by $\sqrt{2}$:

$$s'' = xs - 2s^3,$$

the homogeneous ($\alpha = 0$) Painlevé II equation. So the instanton is a Painlevé II transcendent (the Hastings–McLeod-type solution decaying at $+\infty$).

D.2 (Pohozaev reduction; $c = 5.4439\ldots$, not 2π).

Solution. For the decaying solution of $-g'' + xg = g^3$: multiplying by g and integrating gives $\int g'^2 + \int xg^2 = \int g^4$; multiplying by g' and integrating (a Pohozaev/virial identity) gives a second relation, and together they yield

$$\int g'^2 = \int xg^2 = \tfrac{1}{2}\int g^4 =: c.$$

Hence the variational value $\frac{1}{2}\min K[\psi]^2/\int \psi^4$ collapses to $c = \int g'^2$. Numerically (shooting on the Painlevé II solution, verified to 10^{-13} via the Pohozaev identities) $c = 5.4439\ldots$. A finite-η slope fit overshoots to ≈ 6.22 because the prefactor has not yet decayed; the asymptotic value is the Painlevé II action, and there is *no* reason for it to equal $2\pi \approx 6.283$ — the coincidence is numerical only.

D.3 (Exact TW_2 density and the Monte-Carlo overlay).

Solution. $F_2(s) = \exp\!\big(-\int_s^\infty (x - s)q(x)^2\mathrm{d}x\big)$ with q the Hastings–McLeod solution of $q'' = xq + 2q^3$, $q \sim \mathrm{Ai}$ at $+\infty$; differentiating gives the exact TW_2 density. Monte-Carloing the inner Riccati (1) at $\eta = \sqrt{2}$ ($\beta = 2$) and histogramming Y_{node} with *no* centring or scaling reproduces this density (mean, variance, skewness and the asymmetric tails), confirming Figure 1.

Module E — Excitability and the phase channel

E.1 (Adler bottleneck $\to$ canonical noisy saddle-node).

Solution. Near the SNIC the slow phase obeys the Adler equation $\mathrm{d}\theta = (1-\cos\theta)\mathrm{d}t + \sigma\mathrm{d}W$; at the degenerate saddle $\theta \approx 0$, $1-\cos\theta \approx \theta^2/2$, so $\mathrm{d}\theta = \frac{1}{2}\theta^2\mathrm{d}t + \sigma\mathrm{d}W$. Scale $\theta = \sigma^{2/3}u$, $t = \sigma^{-2/3}\tau$: then $\mathrm{d}\theta = \sigma^{2/3}\mathrm{d}u$, $\frac{1}{2}\theta^2\mathrm{d}t = \frac{1}{2}\sigma^{4/3}u^2\sigma^{-2/3}\mathrm{d}\tau = \frac{1}{2}\sigma^{2/3}u^2\mathrm{d}\tau$, and $\sigma\mathrm{d}W = \sigma\cdot\sigma^{-1/3}\mathrm{d}W_\tau = \sigma^{2/3}\mathrm{d}W_\tau$ (since $\mathrm{d}W \sim \sqrt{\mathrm{d}t} = \sigma^{-1/3}\mathrm{d}W_\tau$). Dividing by $\sigma^{2/3}$:

$$\mathrm{d}u = \tfrac{1}{2}u^2\,\mathrm{d}\tau + \mathrm{d}W_\tau,$$

the parameter-free canonical noisy saddle-node.

E.2 (MFPT quadrature $\Rightarrow$ the 2/3 law).

Solution. The mean first-passage time of $\mathrm{d}u = \frac{1}{2}u^2\mathrm{d}\tau + \mathrm{d}W$ across the bottleneck is the double quadrature $\langle\tau\rangle = 2\iint_{w<u} e^{(w^3-u^3)/3}\mathrm{d}w\,\mathrm{d}u =: 2J$ (Watson's lemma controls the tails, giving convergence). Undoing $t = \sigma^{-2/3}\tau$, the physical rotation rate and phase diffusion inherit the prefactor:

$$\omega = \frac{\pi}{J}\,\sigma^{2/3}\big(1+o(1)\big), \qquad D_\phi \sim \sigma^{2/3},$$

the bifurcation-specific 2/3 exponent of the SNIC channel, $J = \iint_{w<u} e^{(w^3-u^3)/3}$.

E.3 (Fold of cycles gives regular σ^2 diffusion).

Solution. If the cycle dies at a *fold of cycles* the period stays finite at the bifurcation, so the phase-response curve is bounded and the phase obeys an ordinary averaged diffusion $D_\phi = \sigma^2\langle Z^2\rangle \sim \sigma^2$ (regular, no anomalous exponent). The singular $\sigma^{2/3}$ law is special to the *infinite-period* SNIC, where the diverging period $\sim$ log near the bottleneck concentrates the phase noise. Thus the destroying bifurcation selects which channel is singular: amplitude (Tracy–Widom) for the fold of cycles, phase (2/3) for the SNIC.

Module S1 — The shooting characterisation (keystone, §4.3)

S1 (Reconstruct $Y_{\mathrm{node}} \overset{d}{=} -\Lambda_0(\beta) \overset{d}{=} \mathrm{TW}_\beta$).

Solution. *(i) The spectral function.* For $\ell \in \mathbb{R}$ let $G(\ell)$ be the ground-state eigenvalue of

$$M = -\partial_Y^2 + Y - \eta\tilde{\xi} \quad \text{on } [\ell,\infty),\ \text{Dirichlet at } \ell,$$

where $\tilde{\xi}$ is the (reflected) white noise. *(ii) Monotonicity.* By Dirichlet domain monotonicity (F3.2), shrinking the half-line $[\ell,\infty)$ as ℓ increases raises the ground state, so G is strictly increasing; therefore the random first node and the spectral level are linked by the *pathwise* identity

$$\{Y_{\mathrm{node}} \le t\} = \{G(t) \ge 0\}.$$

(The first node of the Cole–Hopf u lies left of t iff the operator on $[t,\infty)$ has no nonpositive eigenvalue, i.e. $G(t) \ge 0$.) *(iii) Removing the random level.* The level t on the right-hand side is now *deterministic*. Shift $Y = t + x$: then $M|_{[t,\infty)} = \big(-\partial_x^2 + x - \eta\hat{\xi}\big) + t$, and by *stationarity and reflection symmetry* of white noise (F2.2) $\hat{\xi} \overset{d}{=} b'$, so $G(t) \overset{d}{=} \Lambda_0(\beta) + t$ with $\Lambda_0(\beta)$ the ground state of the fixed half-line operator $\mathcal{H}_\beta$ on $[0,\infty)$. *(iv) Conclude.*

$$\mathbb{P}(Y_{\mathrm{node}} \le t) = \mathbb{P}(G(t) \ge 0) = \mathbb{P}(\Lambda_0(\beta) \ge -t) = \mathbb{P}(-\Lambda_0(\beta) \le t),$$

hence $Y_{\mathrm{node}} \overset{d}{=} -\Lambda_0(\beta) \overset{d}{=} \mathrm{TW}_\beta$, $\beta = 4/\eta^2$.

Forgetting lemma. Because the canard branch is attracting, the finite-Y_0 swept solution converges (as $Y_0 \to \infty$) to the recessive solution used above (B2.2); this links the abstract Y_{node} to the genuine dynamical peel-off of the trajectory.

Black box vs. proved here. The single imported result is the RRV meaning of $\mathcal{H}_\beta$ and $-\Lambda_0(\beta) \overset{d}{=} \mathrm{TW}_\beta$ (Module C). Everything else — the monotone spectral function G and the collapse of the *random* level to a deterministic one via stationarity/reflection — is proved here; that collapse is the original obstruction the paper removes.

Module S2 — Section-by-section dependency map

S2.

Solution.

Paper element	Supported by
§1 Introduction; eq. (1) inner Riccati	A, F4
§2 Normal form, blow-up, charts; $\eta = \sigma/\sqrt{\varepsilon_2}$	A (B1 for Prop. 8)
§2.3 Two channels; hazard (5)	B2, F4
§3 Main theorems (statements)	C, D
§4.1 Cole–Hopf, Lemma 6	F4, B1
§4.2 Stochastic Airy operator; Sturm fact	C, F3
§4.3 Shooting characterisation (Thm 2)	S1 = A+B+C+F3
§5 Escape probability and rate constant c	B3, D
§6 θ-uniformity, composition, blow-down (Prop. 8)	A, B1–B2
§7 Phase channel; SNIC; 2/3 exponent	E, F4
§8 Numerical validation	C, D (B for Monte-Carlo)
§9 Universality and applications	C (KPZ), E

Optional reproduction (carried out). (1) Monte-Carlo of (1) at $\beta = 1, 2$ overlays the exact TW densities with no fitting (Fig. 1). (2) A random tridiagonal discretisation of $\mathcal{H}_\beta$ matches the ODE-shooting law of Y_{node} at $\beta = 1, 2, 4$ (Fig. 2). (3) The local slope $-\mathrm{d} \log P_{\text{early}}/\mathrm{d}(\eta^{-2})$ descends toward $c = 5.444$; the early-escape tail matches the TW CDF ($\beta = 2$: 0.0298 vs. 0.0306; $\beta = 1$: 0.1644 vs. 0.1681), and the physical law collapses in $\eta = \sigma/\sqrt{\varepsilon_2}$ across a factor four in ε_2. *Beyond the syllabus:* a recrossing analysis shows the TW law is the small-noise edge ($\eta \lesssim 1$); for larger η a static-Kramers boundary $\sigma^2 \sim \Delta U/\log(1/\varepsilon_2)$ takes over.

Part II — "Can you explain it?" (§11 self-test)

1. **Folded limit cycle vs. fold point vs. folded node.** A *fold point* is a saddle–node of *equilibria* on a critical manifold of a 2-D fast–slow system. A *folded node* is a singularity of the *reduced (slow) flow* at a fold curve in a system with ≥ 2 slow variables, generating canards and mixed-mode oscillations. A *folded limit cycle* is the analogue one dimension up in the cyclic direction: a one-parameter family of *limit cycles* (a manifold of cycles) that has a saddle–node of *periodic orbits* — an attracting and a saddle cycle collide — as a slow parameter drifts. The paper perturbs this last object.

2. **Why slow passage makes a canard; deterministic escape a_1.** As the slow variable Y drifts through the fold, the trajectory tracks the attracting branch $R \sim -\sqrt{Y}$ but cannot turn instantly when the branch disappears; it follows the *repelling* continuation for an $O(1)$ inner time — a canard — before escaping. In the inner Airy variable the attracting solution is $R = \mathrm{Ai}'(Y)/\mathrm{Ai}(Y)$, which blows up at the first Airy zero $a_1 \approx -2.3381$: the deterministic peel-off.

3. **What the blow-up accomplishes; why weights $(1, 2, 3)$.** At the fold normal hyperbolicity fails and Fenichel theory breaks down. The geometric blow-up replaces the degenerate point

by a sphere, lifting the dynamics to charts where hyperbolicity is restored and the inner Airy/Riccati problem becomes a regular boundary-value problem. The weights $(1, 2, 3)$ on (r, y, ε_2) are forced by quasi-homogeneous invariance of the fold normal form $r' = r^2 - y$ with slow drift $y' \sim \varepsilon_2$ (A.2): $p_y = 2p_r$, $p_\varepsilon = 3p_r$.

4. **The inner Riccati (1) and $\eta = \sigma/\sqrt{\varepsilon_2}$.** $\mathrm{d}R = (R^2 - Y)\mathrm{d}T + \eta\, \mathrm{d}B$, $Y = Y_0 - T$. It is the universal local model of a noisy slow passage through a quadratic fold. The effective noise $\eta = \sigma/\sqrt{\varepsilon_2}$ arises because the rescaling $r = \varepsilon_2^{1/3} R$, $T = \varepsilon_2^{1/3} t$ amplifies a fixed physical noise σ by $\varepsilon_2^{-1/2}$ (A.1): faster passage (larger ε_2) means relatively weaker noise.

5. **Why no Itô–Stratonovich correction (Lemma 6).** The physical noise is *additive* (constant coefficient), and the Cole–Hopf map $R = -v/u$ is linear in the only noisy variable v, so $R_{vv} = 0$ and the quadratic-variation correction vanishes (B1.1). Itô = Stratonovich here; the Riccati picks up $\eta\, \mathrm{d}B$ with no drift shift.

6. **What Cole–Hopf $R = -u'/u$ achieves; blow-up = node.** It linearises the quadratic Riccati to the linear Schrödinger/Airy equation $u'' = (Y - \eta\xi)u$, turning a blow-up problem into a spectral/first-zero problem. Since $R = -u'/u$, a finite-time divergence of R is exactly a simple zero (node) of u; "escape" = "first node".

7. **The stochastic Airy operator $\mathcal{H}_\beta$ and why its ground state is TW.** $\mathcal{H}_\beta = -\partial_x^2 + x + \frac{2}{\sqrt{\beta}} b'$ on $[0, \infty)$ is the continuum limit of the tridiagonal β-ensemble at the soft edge (RRV); its ground state $-\Lambda_0(\beta)$ is, by construction, the soft-edge fluctuation, i.e. $-\Lambda_0(\beta) \overset{d}{=} \mathrm{TW}_\beta$. The Cole–Hopf of the inner Riccati is exactly this operator with $\eta = 2/\sqrt{\beta}$.

8. **Dictionary $\beta = 4/\eta^2$; Theorem 2 in words.** Matching the noise strengths $\eta = 2/\sqrt{\beta}$ gives $\beta = 4/\eta^2$. Theorem 2: the slow-variable value at canard escape (the peel-off) is Tracy–Widom distributed, $Y_{\mathrm{node}} \overset{d}{=} -\Lambda_0(\beta) \overset{d}{=} \mathrm{TW}_\beta$.

9. **Why the swept-first-node law = the fixed operator's ground state (G).** Define $G(\ell)$ = ground state of the operator on $[\ell, \infty)$. Dirichlet monotonicity makes G strictly increasing, giving the pathwise identity $\{Y_{\mathrm{node}} \le t\} = \{G(t) \ge 0\}$ with t *deterministic*; then stationarity and reflection of white noise give $G(t) \overset{d}{=} \Lambda_0(\beta) + t$, hence $Y_{\mathrm{node}} \overset{d}{=} -\Lambda_0(\beta)$. G is what converts a *random* spectral level (the obstruction) into a deterministic shift.

10. **Early-escape $1 - F_\beta(0)$ vs. hazard $H = \eta^2/4\pi$.** $P_{\mathrm{early}} = \mathbb{P}(Y_{\mathrm{node}} > 0) = \mathbb{P}(\Lambda_0(\beta) < 0) = 1 - F_\beta(0)$ is the probability of peeling off *before* the deterministic fold — the left tail of TW_β, exponentially small ($\sim e^{-c/\eta^2}$). The hazard $H = \eta^2/4\pi$ is the integrated *saddle-crossing* rate, *polynomial* in η, and sets the critical noise $\eta_* = 2\sqrt{\pi}$. They are different observables: a separatrix crossing is not yet a full escape, so $P_{\mathrm{early}} < H$.

11. **Why $c = 5.4439\ldots$ is a Painlevé II action, not 2π.** The LDP rate is $c = \inf\{\frac{1}{2}\int g'^2 : F(g) \le 0\} = \frac{1}{2}\min K[\psi]^2/\int \psi^4$, whose minimiser solves $-g'' + xg = g^3$. Via $g = \sqrt{2}s$ this is Painlevé II $s'' = xs - 2s^3$, so $c = \int g'^2$ is its action; Pohozaev identities give $\int g'^2 = \int xg^2 = \frac{1}{2}\int g^4$ and the value $5.4439\ldots$. The near-coincidence with 2π is numerical; a finite-η fit even overshoots to ≈ 6.22 before settling.

12. **SNIC $\sigma^{2/3}$ vs. fold-of-cycles σ^2 in the phase channel.** A SNIC has *infinite* period at onset; near the bottleneck the Adler equation rescales ($\theta = \sigma^{2/3}u$) to the canonical noisy saddle-node, whose first-passage law gives $\omega, D_\phi \sim \sigma^{2/3}$. A fold of cycles keeps a *finite* period, so the phase response is bounded and phase diffusion is the ordinary $D_\phi \sim \sigma^2$. The destroying bifurcation decides which channel is singular.

13. **Universality claim and physical predictions.** The canard-escape (peel-off) law sits in the Tracy–Widom / KPZ *edge* universality class: any white-noise-perturbed slow passage whose inner equation is the Airy/Riccati canard Cole–Hopf-maps to the stochastic Airy operator, giving TW_β with β read off $\eta = 2/\sqrt{\beta}$. Concrete predictions: peel-off scales $(r, y) \sim (\sigma^{2/3}, \sigma^{4/3})$ and physical critical noise $\sigma_* \propto \sqrt{\varepsilon_2}$ (with geometry factor $G = \sqrt{ac/b}$).